Longbiao CHEN

Academic Curriculum Vitae | Ph.D.

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Research Profile

Research in crowd intelligence, embodied intelligence, spatial computing, ubiquitous computing and domain-specific large language models, with applications in urban governance, public safety, robotic services and healthcare. Ph.D. degrees from Zhejiang University and Sorbonne University.

Education

2015.11-2018.07 Ph.D., Computer Science, Sorbonne University, France. Advisors: Daqing Zhang and Thi-Mai-Trang Nguyen.

2010.09-2016.09 Ph.D., Computer Science and Technology, Zhejiang University, China. Advisor: Gang Pan.

2006.09-2010.07 B.E., Computer Science and Technology, Chu Kochen Honors College, Zhejiang University, China.

Academic and Research Experience

2021.08 Promoted to Associate Professor, Department of Computer Science and Technology, School of Informatics, Xiamen University (former appointment).

2018.08-2021.07 Assistant Professor, Department of Computer Science, School of Informatics, Xiamen University, China.

2016.10-2018.07 Assistant Researcher, Networks and Multimedia Department, Institut Mines-Telecom, France.

Selected Honors and Awards

- 2022: ACM SIGSPATIAL China Rising Star Award.

- 2015 and 2016: ACM UbiComp Honorable Mention Awards.

- 2018: First Prize, Fujian Provincial Science and Technology Progress Award, for multi-source urban transportation sensing and intelligent computing (ranked ninth).

- Huawei Cloud Pioneer Teacher Award.

Recent and Selected Publications

1. Jiedong Yan, Ding He, Junxiang Ji, Guofeng Luo, **Longbiao Chen**. GUIDE: A Conversational LLM Framework for Dynamic Deep Reinforcement Learning Based Building Energy Management. *MobiQuitous 2025 (published online 2026)*, 2026. 10.1007/978-3-032-22500-9_2.

2. Hongjie Fu, Yunqian Li, Guofeng Luo, Ding He, **Longbiao Chen**. FISPA: A LLM-Based Assistance Framework for Fire Investigation. *MobiQuitous 2025 (published online 2026)*, 2026. 10.1007/978-3-032-22500-9_3.

3. Xiaoqin He, Xiaohong Zhong, Jiaru Wang, Kaixuan Zhen, Jinzhun Wu, Boya Tian, **Longbiao Chen**, Haolun Yan, Guorong Lyu. Cleft lip and palate-generative pre-trained transformer: A parameter-efficient, domain-specific language model for cleft lip and palate clinical applications. *Digital Health*, 2026. 10.1177/20552076261416336.

4. Guofeng Luo, Junxiang Ji, Yongyi Wu, Ruixiang Luo, Jiaru Wang, Lijuan Weng, Zhuohan Ye, Chenhui Lu, Dingqi Yang, Cheng Wang, **Longbiao Chen**. UVX-ray: Urban Village Safety Risk Diagnosis Leveraging Multi-Source Urban Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025. 10.1145/3729477.

5. Yayao Hong, Liyue Chen, Leye Wang, Xiuhuai Xie, Guofeng Luo, Cheng Wang, **Longbiao Chen**. STKOpt: Automated Spatio-Temporal Knowledge Optimization for Traffic Prediction. *Proceedings of the ACM Web Conference 2025*, 2025. 10.1145/3696410.3714598.

6. Guofeng Luo, Lijuan Weng, Yunqian Li, Yilu Sun, Yayao Hong, Yongyi Wu, Ruixiang Luo, Leye Wang, Cheng Wang, **Longbiao Chen**. FireExpert: Fire Event Identification and Assessment Leveraging Cross-Domain Knowledge and Large Language Model. *IEEE Transactions on Mobile Computing*, 2025. 10.1109/TMC.2025.3528413.

7. Yuanyuan Zeng, Zhiyu You, Jiayang Guo, Jialin Zhao, Ying Zhou, Jialiang Huang, Xiaowen Lyu, **Longbiao Chen**, Qiyuan Li. Chrombus-XMBD: a graph convolution model predicting 3D-genome from chromatin features. *Briefings in Bioinformatics*, 2025. 10.1093/bib/bbaf183.

8. Yufei Wang, Wenting Zeng, Changzhen Liu, Zhuohan Ye, Jiawei Sun, Junxiang Ji, Zhihan Jiang, Xianyi Yan, Yongyi Wu, Yigao Wang, Dingqi Yang, Leye Wang, Daqing Zhang, Cheng Wang, **Longbiao Chen**. CrowdBot: An Open-Environment Robot Management System for On-Campus Services. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2024. 10.1145/3659601.

9. Yayao Hong, Hang Zhu, Tieqi Shou, Zeyu Wang, Liyue Chen, Leye Wang, Cheng Wang, **Longbiao Chen**. STORM: A Spatio-Temporal Context-Aware Model for Predicting Event-Triggered Abnormal Crowd Traffic. *IEEE Transactions on Intelligent Transportation Systems*, 2024. 10.1109/TITS.2024.3390185.

10. Tieqi Shou, Zhuohan Ye, Yayao Hong, Zhiyuan Wang, Hang Zhu, Zhihan Jiang, Dingqi Yang, Binbin Zhou, Cheng Wang, **Longbiao Chen**. CrowdQ: Predicting the Queue State of Hospital Emergency Department Using Crowdsensing Mobility Data-Driven Models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2023. 10.1145/3610875.

11. **Longbiao Chen**, Xin He, Xiantao Zhao, Han Li, Yunyi Huang, Binbin Zhou, Wei Chen, Yongchuan Li, Chenglu Wen, Cheng Wang. GoComfort: Comfortable Navigation for Autonomous Vehicles Leveraging High-Precision Road Damage Crowdsensing. *IEEE Transactions on Mobile Computing*, 2023. 10.1109/TMC.2022.3198089.

12. **Longbiao Chen**, Daqing Zhang, Gang Pan, Xiaojuan Ma, Dingqi Yang, Kostadin Kushlev, Wangsheng Zhang, Shijian Li. Bike sharing station placement leveraging heterogeneous urban open data. *2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2015. 10.1145/2750858.2804291.

Full publication list: longbiao.crowdsensing.cn/publications/

Academic Service

- Editor-in-Chief, Big Data Research; editorial board service for Vehicles, Frontiers in Sustainable Cities and JILSA.

- Executive committee member, ACM SIGSPATIAL China Chapter; executive committee member, CCF Ubiquitous Computing Committee.

- CCF YOCSEF Xiamen: Chair (2021), Vice Chair (2020), Academic Secretary (2019).

- Social Activity Chair, ACM UbiComp 2021; Industry Chair, AIoTSys 2025.

- Member, BRICS Young Scientist Forum and moderator of its AI session (2022); program committee service for AAAI and IJCAI.

Edited Volume

Co-editor: Spatial Data and Intelligence, SpatialDI 2025 proceedings, LNCS 15838, Springer. DOI.

Selected Research Grants as Principal Investigator

2022-2025 NSFC General Program, Principal Investigator.

2019-2021 NSFC Young Scientists Fund, Principal Investigator.

2019-2021 National defense basic research project, Principal Investigator.

2020-2021 National high-resolution Earth observation project, Principal Investigator.

2018-2021 Natural Science Foundation of Fujian Province, General Program, Principal Investigator.

2020-2021 Xiamen industry-university-research collaborative innovation project, Principal Investigator.

Teaching and Mentoring

Teaching includes software engineering, pattern recognition, software architecture and application development, human-computer interaction, and data mining. Led a Ministry of Education “Smart Base” industry-education collaborative program and supervised student research and innovation competitions.

Selected Patents

- Deep-learning-based venue flow control method, system and readable medium. 2022112269437

- Robot task management system based on deep reinforcement learning. 202411116566.0

- Deep-learning-based urban-village risk assessment and risk-factor localization. 202210493040.9

- Deep-learning-based urban-village identification and population estimation. 202010873246.5